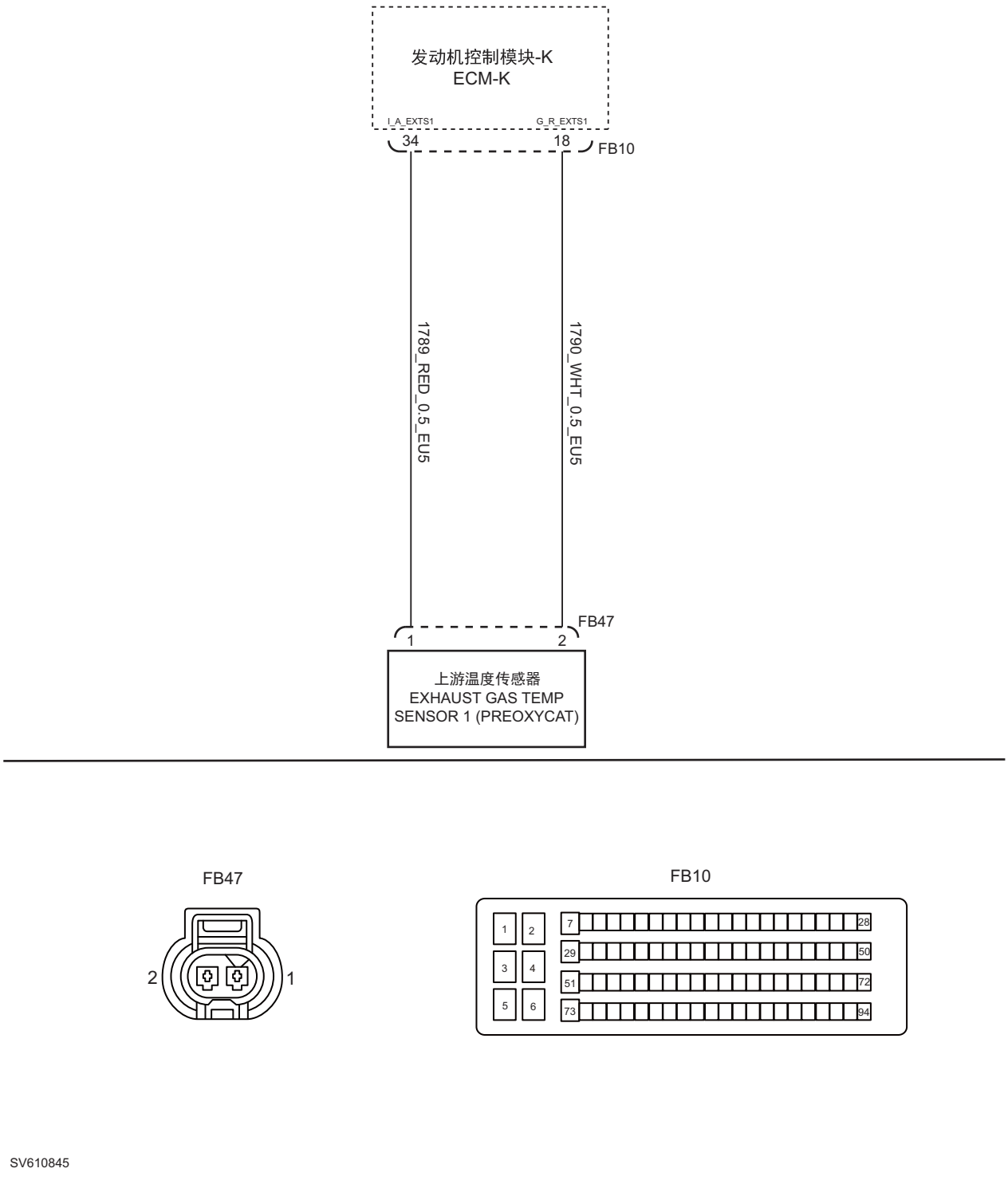
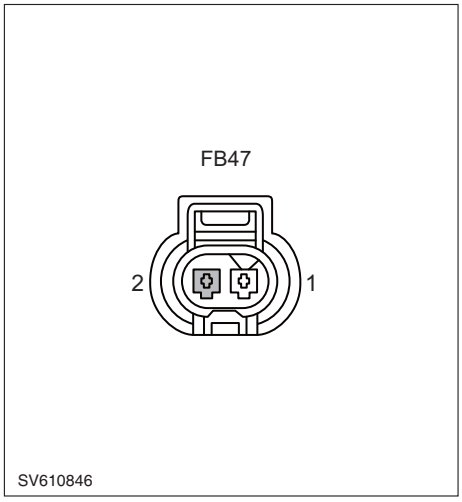
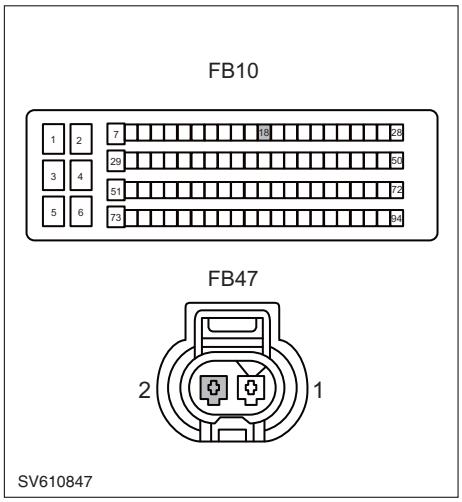
P0425

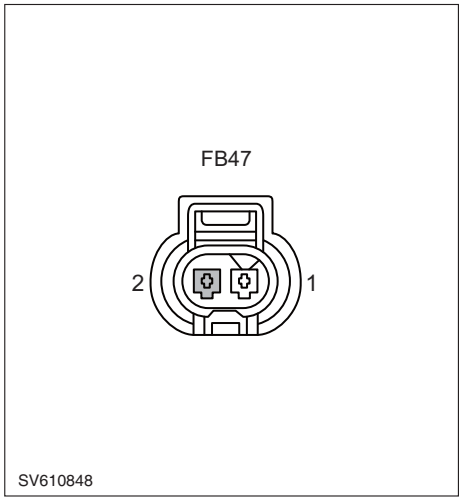
Description of fault diagnosis codes
P0425-11:Diagnostic fault check for SRC low in Oxidation Catalyst upstream temperature
P0425-12:Diagnostic fault check for SRC high in Oxidation Catalyst upstream temperature
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • Fault in upstream temperature sensor
Description of circuit diagram
The discharge temperature sensor is used to detect the discharge temperature and feed back the voltage signal to ECU. ECU will detect and control the discharge temperature during DPF regeneration according to two discharge temperature signals.

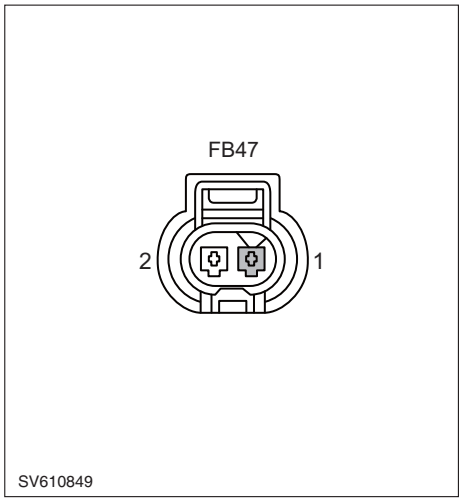
Circuit diagram



Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check if there is any damage, bad contact, aging and looseness phenomenon on harness plug of engine control module and upstream temperature sensor. 4 Check whether the result is normal?
		No → Repair or replace the faulted parts.
Yes ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the engine control system with diagnostic apparatus. 4 Is there any other fault code except for P0425?
		Yes → Reference:DTC Summary List (EU5).
No ↓		
Step 3	Read the data stream of upstream temperature sensor with diagnostic apparatus.	
		1 Connect the failure diagnosis instrument to the diagnosis interface. 2 Read the actual value of upstream temperature sensor signal with diagnostic apparatus. **Standard value: 0.35~0.50V (0°C)** **Standard value: 0.40~0.53V (25°C)** **Standard value: 0.45~0.55V (50°C)** **Standard value: 0.50~0.65V (100°C)** 3 Check whether the result is normal?
		Yes → Complete the check and confirm whether the system works normally nor not.
No ↓		

Step 4	Check the signal power circuit of upstream temperature sensor.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of upstream temperature sensor. 3 Turn the ignition switch to ON position. 4 Measure the voltage between the harness plug of upstream temperature sensor and ground. Measuring circuit: the voltage between terminal 2 of FB47 plug and ground. **Standard value: 4.8~5.2V** 5 Check whether the result is normal?
	Yes → Go to Step 7.
No ↓	
Step 5	Check the signal power supply circuit (open circuit) of upstream temperature sensor.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Disconnect the harness plug of upstream temperature sensor. 4 Measure the resistance between harness plug of engine control module and harness plug of upstream temperature sensor. Measuring circuit: the resistance between No.18 terminal of FB10 plug and terminal 2 of FB47 plug. **Standard value: ≈ 0Ω** 5 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 6	Check the signal power circuit (short circuit) of upstream temperature sensor.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Disconnect the harness plug of upstream temperature sensor. 4 Measure the resistance between the harness plug terminal of upstream temperature sensor and ground. Measuring circuit: the resistance between terminal 2 of FB47 plug and ground. **Standard value: ∞** 5 Measure the voltage between the harness plug terminal of upstream temperature sensor and ground. Measuring circuit: the voltage between terminal 2 of FB47 plug and ground. **Standard value: ≈ 0V** 6 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 7	Check the signal ground circuit of upstream temperature sensor.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Disconnect the harness plug of upstream temperature sensor. 4 Measure the resistance between the harness plug terminal of upstream temperature sensor and ground. Measuring circuit: the resistance between terminal 1 of FB47 harness plug and ground. **Standard value: ≈ 0Ω** 5 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

<table>
<tr><td>Step 8</td><td colspan="2">Check the upstream temperature sensor (component inspection).</td></tr>
<tr><td colspan="2"> 
</td><td>1 Turn the ignition switch to OFF.
2 Disconnect the harness plug of upstream temperature sensor.
3 Measure the resistance between the harness plug terminals of upstream temperature sensor.
Measuring circuit: the resistance between terminal 1 and terminal 2 of FB47 harness plug.
Standard value: ≈ 201Ω(at 0°)
Standard value: ≈ 220Ω(at 25°)
Standard value: ≈ 239Ω(at 50°)
Standard value: ≈ 276Ω(at 100°)
4 Check whether the result is normal?</td></tr>
<tr><td colspan="2"></td><td>No → Replace upstream temperature sensor.</td></tr>
<tr><td colspan="3">Next ↓</td></tr>
<tr><td>Step 9</td><td colspan="2">Complete the check and confirm whether the system works normally nor not.</td></tr>
</table>